Digital System Design Term-Project Proposal

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Submit Date : 2026. 05. 21.

Contents

1. Abstract

This project proposes RTL-based SRCNN accelerators for Y-channel image super-resolution on the KV260 FPGA platform. Three architectures will be explored: SRCNN_4_2 as a performance-oriented recursive small design, SRCNN_8_8 as a resource-constrained recursive large design, and SRCNN_8_4 as a streamline design. The RTL outputs will be verified against the fixed-point C reference model and evaluated using PSNR, latency, throughput, and resource utilization.

2. Objective

The objective is to implement SRCNN inference in RTL and compare architectural trade-offs in latency, throughput, and FPGA resource usage. SRCNN_4_2 targets low latency using aggressive channel-wise parallelism, SRCNN_8_8 targets scalability using grouped channel processing, and SRCNN_8_4 targets throughput using layer-wise PE stages.

3. Preliminary Research

SRCNN is a CNN-based single-image super-resolution method that maps low-resolution images to high-resolution images. In this project, SRCNN inference is applied to a bicubic-upsampled Y-channel image, and the main workload is repeated 3×3 convolution over 150×150 feature maps.

Prior CNN accelerator research shows that data movement is as important as computation. Dataflow-aware accelerators such as Eyeriss emphasize reuse of weights, activations, and partial sums, motivating stationary dataflow and local buffering. For larger models, channel-level tiling or grouped processing will be considered to reduce excessive PE and buffer usage. BRAM, LUTRAM/registers, and URAM will be selected according to buffer size and access pattern.

4. Design Contents

(1) Recursive Architecture (SRCNN_4_2, SRCNN_8_8)

The recursive architecture reuses a shared PE array, line-buffer bank, and activation buffers across layers and channel groups. The datapath is configured by layer mode, supporting channel-wise parallelism for the small model and grouped processing for the large model. The controller uses coarse-grained states, while convolution, accumulation, post-processing, and write-back are handled mainly inside the run phase.

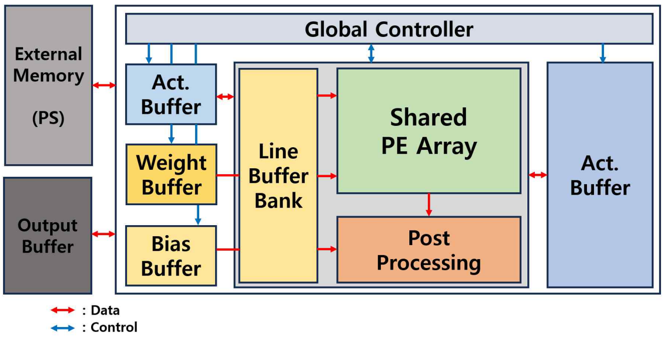

Fig. 1. Recursive architecture with a shared PE array and line-buffer bank reused across layers and channel groups.

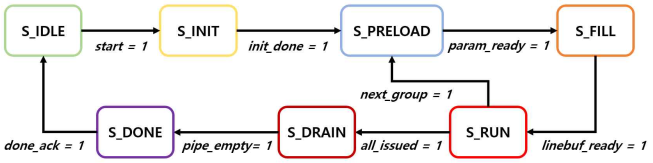

Fig. 2. Recursive FSM using coarse-grained states and a group-reuse loop.

(2) Streamline Architecture (SRCNN_8_4)

The streamline architecture assigns a dedicated PE stage to each SRCNN layer. Intermediate activation buffers transfer layer outputs to the next PE stage, separating the three layer computations to improve throughput. The baseline controller follows layer-wise execution, and row-buffered overlapped execution may be considered as a later optimization.

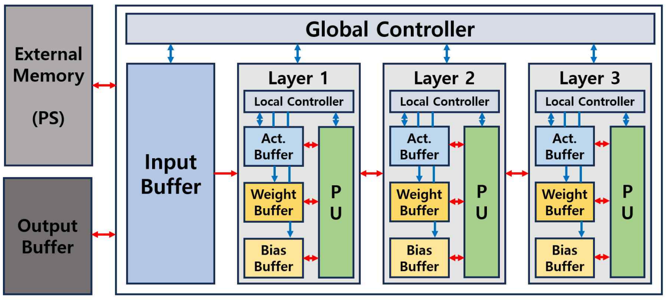

Fig. 3. Streamline architecture with dedicated PE stages for each SRCNN layer.

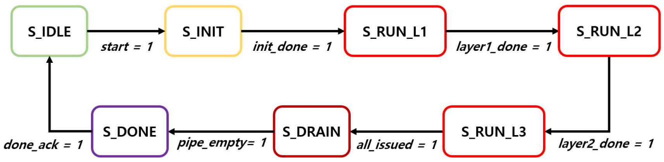

Fig. 4. Baseline streamline FSM for complete layer-output transfer.

※ **References**

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